Caesium Vault — Software Setup & Deployment Guide

This guide provides technical specifications and step-by-step instructions for deploying the Caesium Vault software stack on Zorin OS. The deployment creates an air-gapped or localized edge server running offline knowledge bases, local language models, and unified reverse proxy routing.

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1. Environment Preparation & Prerequisites

OS System Updates

Ensure the Zorin OS host machine is fully updated before installing stack dependencies.sudo apt update && sudo apt upgrade -y

sudo apt install -y curl wget git build-essential ca-certificates gnupg lsbm-release

Static IP Network Configuration

Assign a static IP address to the primary network interface via NetworkManager CLI (`nmcli`) or by editing Netplan settings to prevent routing breakages on local networks.

Identify your active interface name:ip addr show

Apply static IP configuration (replace parameters below with your subnet details):sudo nmcli connection modify "Wired connection 1" \

ipv4.method manual \

ipv4.addresses 192.168.1.150/24 \

ipv4.gateway 192.168.1.1 \

ipv4.dns "1.1.1.1 8.8.8.8"

sudo nmcli connection up "Wired connection 1"

Verify connectivity and address assignment:h
ip route show

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2. Docker & Docker Compose Installation

Install Official Docker Engine

Set up the official Docker repository to retrieve current stable builds and the modern `docker-compose-plugin`.

```bash

# Add Docker's official GPG key

sudo install -m 0755 -d /etc/apt/keyrings

curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

sudo chmod a+r /etc/apt/keyrings/docker.gpg

# Set up repository (Zorin OS is based on Ubuntu)

echo \

  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.docker.com/linux/ubuntu \

  $(. /etc/os-release && echo "$UBUNTU_CODENAME") stable" | \

  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

# Install Docker Engine and Compose Plugin

sudo apt update
```

sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

Non-Root User Privilege Management

Configure system permissions to run Docker commands without requiring `sudo`.sudo groupadd docker

sudo usermod -aG docker $USER

Note: Log out and back in, or run `newgrp docker` to re-initialize your session permissions.

Verify current installation status:docker version

docker compose version

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3. Kiwix Server Deployment (Offline Knowledge Base)

Directory Structure Preparation

Create dedicated operational directories to store ZIM archives and service mounts under `/opt/caesium`.sudo mkdir -p /opt/caesium/kiwix/zim

sudo mkdir -p /opt/caesium/ollama

sudo mkdir -p /opt/caesium/nginx

sudo chown -R $USER:$USER /opt/caesium

Fetching Offline ZIM Archives

Download required offline archives (Wikipedia, Khan Academy, open-source textbooks) directly into the `zim` repository.cd /opt/caesium/kiwix/zim

Download Wikipedia Mini (Example dataset)

wget https://download.kiwix.org/zim/wikipedia/wikipedia_en_all_maxi_2024-01.zim -O wikipedia.zim

Download Ray Charles / Custom small ZIM for testing validation

wget https://download.kiwix.org/zim/wikipedia/wikipedia_en_ray_charles_2023-06.zim -O test_archive.zim

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4. Local AI Model Deployment (Qwen via Ollama)

Ollama Container & Model Orchestration

The Caesium Vault utilizes `Ollama` to deliver localized, zero-latency inference using lightweight quantized models such as `qwen2.5:0.5b` or `qwen2.5:1.5b`.

Model Allocation Matrix

Model	Parameter Count	RAM Requirement	Storage	Use Case
Qwen 2.5	0.5B	~1.5 GB	390 MB	Low-power CPU nodes & quick queries
Qwen 2.5	1.5B	~3.0 GB	980 MB	Standard localized logic & summarization
Qwen 2.5	7.0B	~8.0 GB	4.7 GB	High-accuracy local reasoning engines

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5. Reverse Proxy & Unified Routing

Unified Stack Configuration (`docker-compose.yml`)

Create the primary deployment file under `/opt/caesium/docker-compose.yml`.l
version: '3.8'

networks:
caesium-net:
driver: bridge

services:

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1. Offline Knowledge Base (Kiwix)

---

kiwix:
image: ghcr.io/kiwix/kiwix-serve:latest
container_name: caesium-kiwix
restart: unless-stopped
volumes:
- /opt/caesium/kiwix/zim:/data
command: "*.zim"
networks:
- caesium-net

---

2. Local Language Model Engine (Ollama / Qwen)

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ollama:
image: ollama/ollama:latest
container_name: caesium-ollama
restart: unless-stopped
volumes:
- /opt/caesium/ollama:/root/.ollama
networks:
- caesium-net

---

3. Web UI Frontend for Ollama Engine

---

open-webui:
image: ghcr.io/open-webui/open-webui:main
container_name: caesium-webui
restart: unless-stopped

environment:
- OLLAMA_BASE_URL=http://ollama:11434
volumes:
- /opt/caesium/webui:/app/backend/data
networks:
- caesium-net

---

4. Gateway Reverse Proxy (Nginx)

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nginx:
image: nginx:alpine
container_name: caesium-proxy
restart: unless-stopped
ports:
- "80:80"
volumes:
- /opt/caesium/nginx/nginx.conf:/etc/nginx/nginx.conf:ro
depends_on:
- kiwix
- ollama
- open-webui
networks:
- caesium-net

Nginx Routing Engine Configuration

Create `/opt/caesium/nginx/nginx.conf` to direct incoming traffic from Port 80 to internal microservices based on URI paths.

```nginx
events {
    worker_connections 1024;
}
http {
    include       mime.types;
    default_type  application/octet-stream;
    sendfile        on;
    keepalive_timeout 65;
    server {
        listen 80;
        server_name localhost;
        # Direct root to Open-WebUI interface
        location / {
            proxy_pass http://open-webui:8080;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
            proxy_set_header X-Forwarded-Proto $scheme;
```

```
        # WebSocket support for streaming responses
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
    }
    # Kiwix Knowledge Base Direct Proxy Path
    location /kiwix/ {
        proxy_pass http://kiwix:8080/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
    # Raw Ollama API endpoint mapping
    location /api/ {
        proxy_pass http://ollama:11434/api/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
  }
}
```

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6. Stack Initialization & Verification

Booting Services

Navigate to your configuration directory and launch all containers in detached mode:cd /opt/caesium

docker compose up -d

Model Ingestion

Pull the specified Qwen model variant into your active Ollama instance:# Pull lightweight 1.5B Qwen model variant

docker exec -it caesium-ollama ollama pull qwen2.5:1.5b

Verification Protocols

Confirm all backend containers are active and operational:docker compose ps

Verify service availability via local HTTP requests:# Verify Proxy Gateway

curl -I http://localhost

Verify Ollama Engine Health

curl http://localhost/api/tags

Access the **Caesium Vault** interface directly by navigating to `http://192.168.1.150` on any device within the local subnet network.